**Assignment No. 12**

**Title**

Creating a Version Control Account on GitHub and Using Git Commands to Create a Repository and Push Code

**Problem Statement**

Create an account on GitHub and use Git commands to:

- Initialize a local repository
- Add project files
- Commit changes
- Connect to a remote GitHub repository
- Push code to GitHub

**Objectives**

1. To understand the concept of version control systems.
2. To learn basic Git commands for managing repositories.
3. To create and manage repositories on GitHub.

**Theory**

**1. Version Control System (VCS)**

A **Version Control System (VCS)** is a tool that helps developers manage changes to source code over time.

**Types of VCS:**

1. **Centralized VCS (CVCS)**
   - Example: SVN
   - Single central server
2. **Distributed VCS (DVCS)**
   - Example: Git
   - Each user has a complete copy of repository

**2. Introduction to Git**

Git

Git is a **distributed version control system** used to track changes in source code.

**Features of Git:**

- Fast and efficient
- Distributed system
- Branching and merging
- Data integrity

Git stores data as **snapshots**, not changes.

**3. Introduction to GitHub**

GitHub

GitHub is a cloud-based platform used to host Git repositories.

**Features:**

- Code hosting
- Collaboration
- Version tracking
- Issue tracking
- Pull requests

**4. Git Architecture**

Git has three main areas:

| **Area** | **Description** |
|---|---|
| Working Directory | Files you are working on |
| Staging Area | Files ready to commit |

| | |
|---|---|
| Repository | Saved versions of files |

**5. Git Installation**

Git can be installed from official website:

Git

After installation, configure Git:

git config --global user.name "Your Name"

git config --global user.email "your@email.com"

**6. Creating GitHub Account**

Steps:

1. Visit GitHub
2. Click **Sign Up**
3. Enter email, username, password
4. Verify account

**7. Repository in Git**

A **repository (repo)** is a folder that contains project files and version history.

Types:

- Local repository
- Remote repository

**8. Basic Git Commands**

**1. Initialize Repository**

git init

Creates a new Git repository.

**2. Add Files to Staging Area**

git add .

Adds all files.

**3. Commit Changes**

git commit -m "Initial commit"

Saves changes permanently.

**4. Check Status**

git status

Shows file status.

**5. Connect to Remote Repository**

git remote add origin https://github.com/username/repo-name.git

**6. Push Code to GitHub**

git push -u origin main

Uploads code to GitHub.

**9. Branch Concept**

A **branch** is a separate line of development.

Default branch: main

**10. Workflow of Git**

1. Modify files
2. Add files
3. Commit changes
4. Push to GitHub

**11. Importance of Version Control**

- Tracks history
- Enables collaboration
- Prevents data loss
- Supports team development

**Practical Implementation Steps**

**Step 1: Create Project Folder**

mkdir myproject

cd myproject

**Step 2: Initialize Git**

git init

**Step 3: Create File**

echo "Hello GitHub" > index.html

**Step 4: Add File**

git add .

**Step 5: Commit File**

git commit -m "First commit"

**Step 6: Create Repository on GitHub**

- Login to GitHub
- Click **New Repository**
- Enter repository name

**Step 7: Connect Local to Remote**

git remote add origin https://github.com/username/myproject.git

**Step 8: Push Code**

git push -u origin main

**Conclusion**

In this experiment, we successfully created a GitHub account and used Git commands to manage a repository. The local repository was initialized, files were added and committed, and finally pushed to GitHub. This experiment demonstrates the importance of version control in software development and provides essential skills for collaborative coding and project management.

**Viva Questions**

1. What is version control system?
2. What is Git?
3. What is GitHub?
4. Difference between Git and GitHub?
5. What is repository?
6. What is staging area?
7. What is commit?
8. What is branch?
9. What is push and pull?
10. Why is version control important?